

Beyond Ranks

Inequality in the Measurement of Mobility

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Mobility and Inequality are Intertwined

- Many mobility measures focus on individuals' *relative positions*:
 - Changes in income ranks (Bartholomew 1973; Shorrocks 1978)
 - Rank correlations (Schiller 1977; Chetty, Hendren, Kline, Saez, and Turner 2014; Fagereng, Mogstad, and Rønning 2021, many others!)

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 - Same rank movement can represent very different material changes
 - Invariant to all monotone transformations of income (including progressive taxation)
- Purely cardinal measures miss the role of *relative position* (Galton 1886; Hart 1983; Solon 1992; Zimmerman 1992; Hart–Shorrocks 1993; Fields–Ok 1996; Ray Genicot 2023; Davis–Mazumder 2024)

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Our objective: Measure mobility using both position and material income differences

Measuring Inequality

The Lorenz curve encodes the relative income distribution (Atkinson 1970; Sen 1973)

- Economy with income vector $Y = (y_1, \dots, y_N)$ and mean income μ ,

$$r(y, Y) \equiv \frac{1}{N} \sum_{j=1}^N \mathbb{1}\{y_j \leq y\}, \quad L(r, Y) \equiv \sum_{\{j: r_j \leq r\}} \frac{y_j}{N\mu}.$$

- $L_i = L(r_i, Y)$ is the cumulative income share held by dynasties no richer than i
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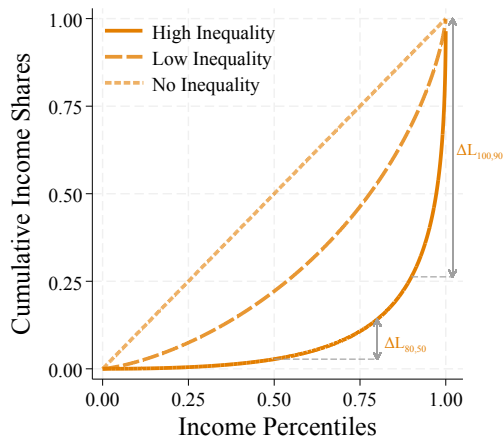
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Lorenz mobility: Observe income over time, Y^P and Y^K , with individual mobility

$$m_i = L_i^K - L_i^P$$

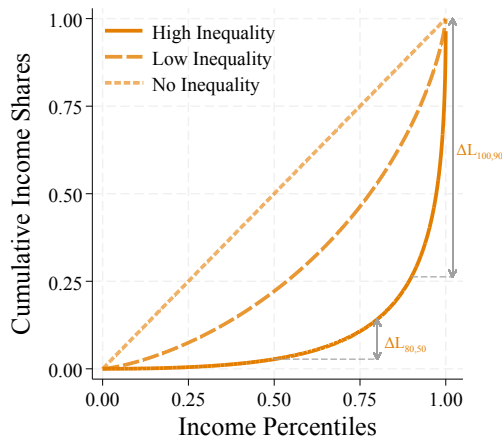
Positional Mobility in Unequal Economies



In unequal economies, income distributions compress at the bottom and spread at the top.

Lorenz ordinates capture this relationship:

Positional Mobility in Unequal Economies



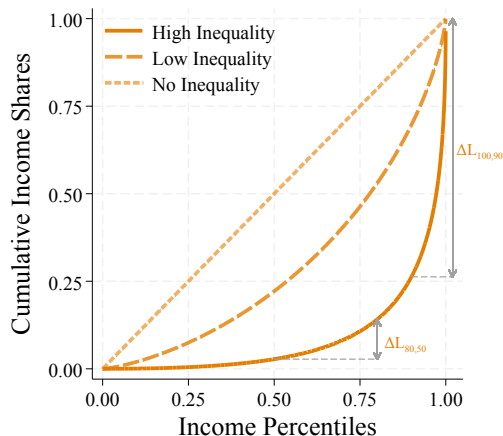
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 1. When income goes up Lorenz ordinate goes up (holding constant income distribution)
 2. Evaluate economy-wide changes in income in the light of **changes in inequality**
- Ranks can miss both cardinal and distributional changes
 - No rank changes when comparing *co-monotone* income distributions
 - Equivalent rank mobility in market vs disposable income under progressive taxation

Inequality, Redistribution, and Mobility

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- Look at the effect of **Pigou-Dalton transfers**
 - Redistribute income to reduce inequality
 - Take dynasties h and j with $y_h < y_j$ and transfer $\delta > 0$ from j to h with $y_h + \delta \leq y_j - \delta$
 - Respect order between h and j but can generate other positional changes

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- Contrast with **income swaps**
 - Change dynasties' positions without changing inequality

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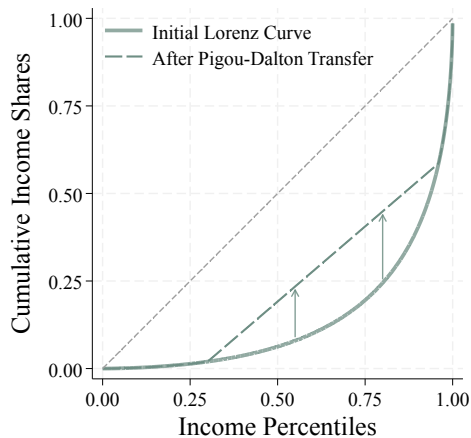
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Movements towards equality increase upward mobility

Redistribution: Pigou-Dalton Transfers

► Proportional transfers



Reducing inequality raises the Lorenz curve

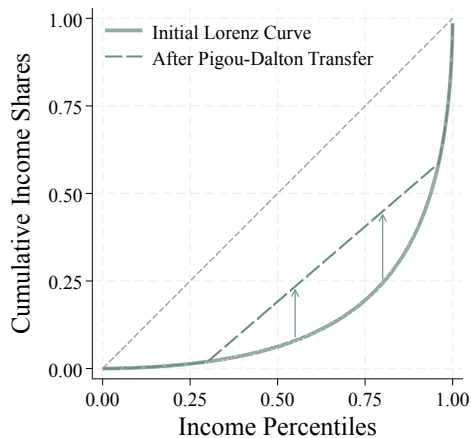
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- Upward mobility if rank is preserved
- Transfers can also change positions!

Net result depends on **positional mobility** ...

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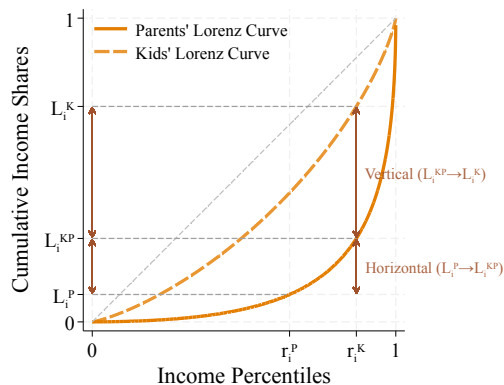
Mobility necessarily higher for all dynasties relative to equivalent income swaps

- Upward movers move farther up
- Downward movers fall by less

Disentangling Changes in Inequality and Changes in Position

Vertical and Horizontal Mobility

$$L_i^K - L_i^P = \underbrace{L_i^K - L_i^{KP}}_{\text{Vertical}} + \underbrace{L_i^{KP} - L_i^P}_{\text{Horizontal}}.$$



Horizontal Mobility:

Change in rank holding inequality constant

- Movement along the parents' Lorenz curve

Vertical Mobility:

Change in inequality holding position fixed

- Movement between Lorenz curves

Co-monotone generations have only vertical mobility;
Identical distributions have only horizontal mobility.

Aggregating Mobility

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Dynastic mobility:

[► details](#)

Signed mobility: $m_i = L_i^K - L_i^P$,

Symmetric mobility: $m_i = |L_i^K - L_i^P|$

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$$\mathcal{M}_N = \left(\frac{1}{N} \sum_{i=1}^N m_i^\gamma \right)^{1/\gamma} = \underbrace{\bar{m}}_{\text{Average Mobility}} \times \underbrace{\mathcal{C}_\gamma(M)}_{\text{Mobility Concentration}}$$

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$$\text{Signed mobility: } m_i = L_i^K - L_i^P, \quad \text{Symmetric mobility: } m_i = \left| L_i^K - L_i^P \right|$$

Mobility Concentration: Curvature γ controls role of concentration

$$\mathcal{C}_\gamma(M) = \exp \left(\frac{\gamma - 1}{\gamma} D_\gamma(M) \right).$$

$D_\gamma(M)$ is the Rényi entropy measure (dispersion relative to uniform distribution)

$\gamma < 1$: penalizes concentrated mobility $\gamma > 1$: favors large, concentrated movements

Average Mobility and Inequality: Signed Mobility

Net Mobility: $\bar{m} \equiv \frac{1}{N} \sum_{i=1}^N (L_i^K - L_i^P) = \frac{1}{2} \underbrace{(\mathcal{G}(Y^P) - \mathcal{G}(Y^K))}_{\text{Change in Gini Coefficient}}$

- Decreases in inequality are increases in (net) mobility
- Completes the mobility-inequality link (even when the Lorenz curves of Y^K and Y^P intersect)
- Net mobility is *vertical mobility*; horizontal mobility is zero-sum

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Symmetric mobility: Additionally captures upward/downward mobility

► details

Analytical Example: Pareto Income and HSV Taxes

Income Distribution

$$y^G \sim \text{Pareto}(\alpha^G)$$

Lorenz Curve

$$L^G(r) = 1 - (1 - r)^{1-1/\alpha^G}$$

Gini Coefficient

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Disposable Income

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► Symmetric mobility

- Children's progressivity raises mobility; Parents' progressivity lowers it.
- Common progressivity preserves mobility sign and attenuates magnitude

Bounding Mobility

Maximal Average Mobility

Objective: Express realized mobility as a fraction of *attainable* mobility.

- Makes different mobility measures comparable

Maximal net mobility

$$\bar{m} = \frac{1}{2} \left[\mathcal{G}(Y^P) - \mathcal{G}(Y^K) \right] \longrightarrow \sup \bar{m} = \frac{1}{2}$$

- Maximal attained going from full concentration of income to full equality
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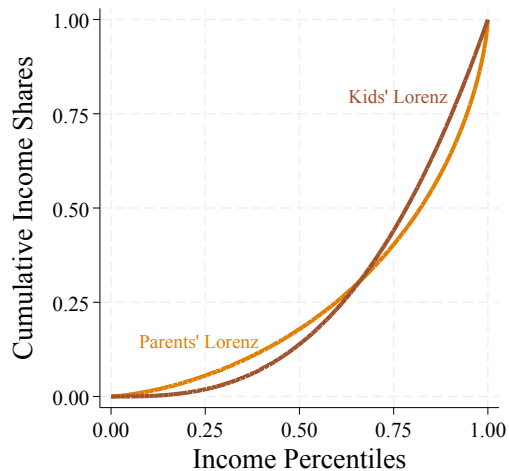
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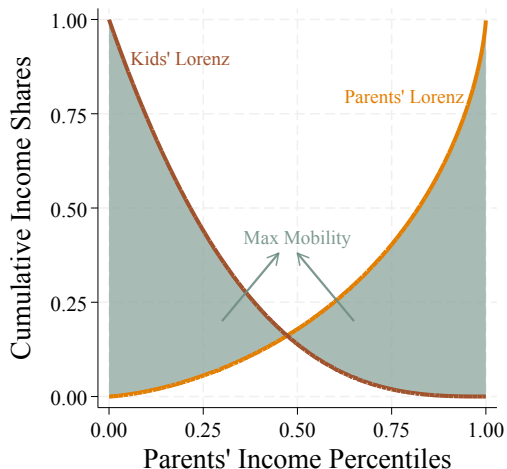
Maximal Symmetric mobility: Rank-dependence + Shape of Lorenz curves

Maximal Symmetric Mobility



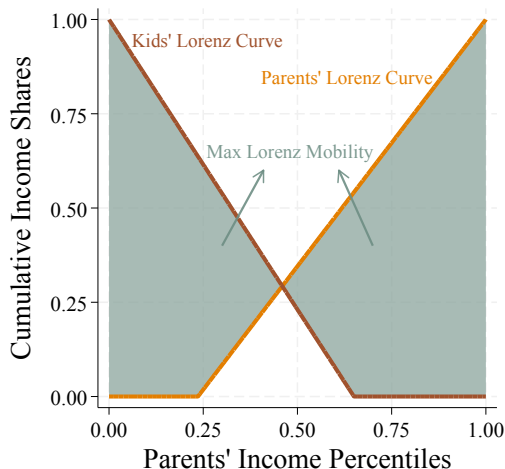
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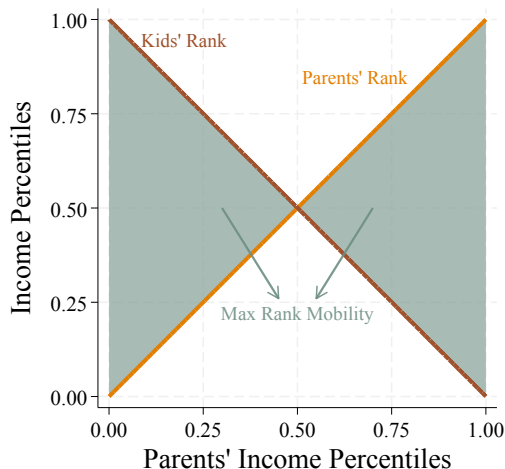
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Maximal Rank Mobility



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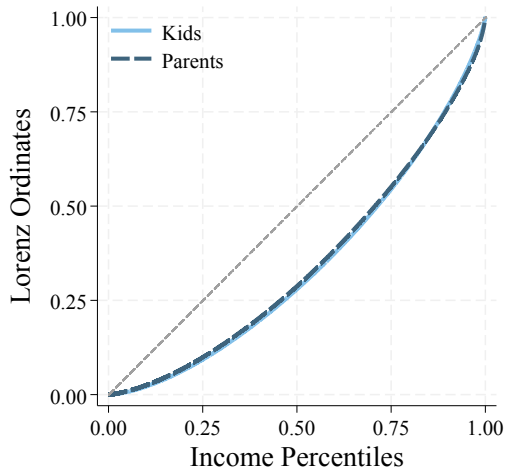
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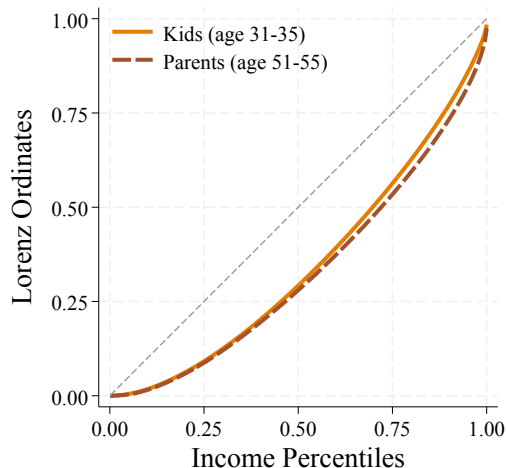
Measuring Intergenerational Mobility in the US and Norway

United States



Linked NLS66–NLSY79 parents and children
(Davis and Mazumder, 2024)

Norway



Linked parent-child tax records
Norwegian administrative data

Signed Mobility and Inequality

$$\mathcal{M}_N = \left(\frac{1}{N} \sum_{i=1}^N (m_i)^\gamma \right)^{1/\gamma}$$

- US inequality $\uparrow$: Mobility < 0
- NOR inequality $\downarrow$: Mobility > 0

		Signed mobility		
		$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$
US	Lorenz	−0.005	−0.000	−0.051
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ranks and Lorenz ordinates
- **Taxes** increase mobility in NOR

Symmetric Mobility: Lorenz mobility is lower than rank mobility

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		Symmetric mobility		
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US	Lorenz	0.26	0.21	0.34
	Rank	0.28	0.23	0.35
Norway	Lorenz	0.28	0.23	0.35
	Rank	0.30	0.25	0.38

- Lorenz mobility is lower for every γ in both countries
- Relative to attainable mobility:
 - US: Lorenz 44%, ranks 56%
 - Norway: Lorenz 47%, ranks 60%
- Lorenz $\approx$ 10pp lower than rank
- Consistent with rank changes in flatter portion of Lorenz curves

Intergenerational Persistence: Similar Results

Galtonian Regression: $x_i^K = \alpha_x + \beta_x x_i^P + u_i$ Mobility: $1 - \hat{\beta}_x$

	Lorenz	Rank	Log-Lorenz	Log-income
US	0.26	0.23	0.27	0.30
Norway	0.18	0.16	0.10	0.11

- Greater persistence (lower mobility) in the US under every measure
- Rank implies somewhat more mobility than Lorenz measures (income somewhat less)
- Ultimately estimates are quantitatively close (McGee 2026)

► Binned scatter plots

Conclusions

Lorenz mobility combines

1. Ordinal information of ranks
2. Cardinal information of income

Provides a measure of mobility that responds to changes in income and inequality

- Inequality determines how positional movements translate into mobility
- Net Lorenz mobility is the change in the Gini coefficient and is panel independent
- Distinguishes positional reshuffling and captures changes in inequality and taxation

Appendix

Lorenz Ordinates as Aspirational Status

Let status depend on own income and the income distribution: $s(y, Y)$.

- **Monotonicity:** status rises with own income
- **Growth independence:** common rescaling does not change status
- **Aspirational status:** status rises as a dynasty approaches those above and moves away from those below
- **Anonymity:** only cumulative income above and below matters, not identities

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$$s(y, Y) = \psi(r(y, Y), L(y, Y)),$$

where ψ is increasing in rank and strictly increasing in the Lorenz ordinate.

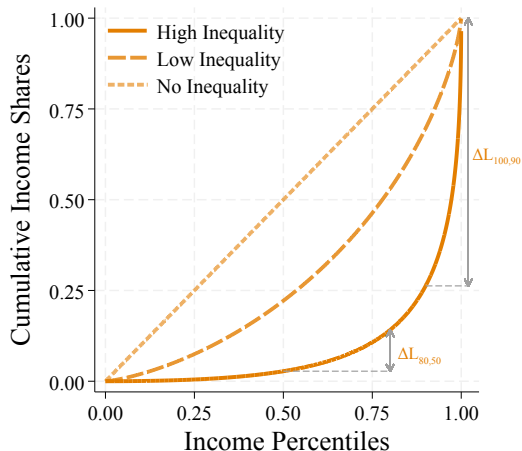
Status Units

Requiring status to move linearly along rays connecting co-monotone distributions gives

$$s(y_i, Y) = \Omega(r_i) [\lambda L_i + (1 - \lambda) \Xi(r_i)], \quad \lambda \in (0, 1].$$

- Lorenz ordinates provide the cardinal content of status
- Increasing rank weights $\Omega(r)$ allow status to depend on position
- $\lambda = 1$ and $\Omega(r) = 1$ give Lorenz ordinates
- Ranks arise as the limiting case $\lambda \rightarrow 0$, $\Omega(r) = 1$, and $\Xi(r) = r$

Redistribution: Proportional Transfers



$$Y^\alpha = \alpha Y + (1 - \alpha)\mu \mathbb{1}, \quad \alpha \in [0, 1].$$

Ranks are unchanged, but

$$L_i^\alpha = \alpha L_i + (1 - \alpha)r_i \geq L_i.$$

- Every dynasty's Lorenz ordinate weakly rises
- Above-average dynasties can lose income while moving upward in Lorenz status
- Redistribution moves the entire curve toward ranks

Signed Mobility Aggregation

[◀ Back](#)

$$\mathcal{M}_N = \text{sign} \left(\frac{1}{N} \sum_{i=1}^N \text{sign}(m_i) |m_i|^\gamma \right) \left| \frac{1}{N} \sum_{i=1}^N \text{sign}(m_i) |m_i|^\gamma \right|^{1/\gamma}.$$

Upward and downward mobility

$$\mathcal{U} \equiv \frac{1}{N} \sum_{i=1}^N |m_i|^+, \quad \mathcal{D} \equiv \frac{1}{N} \sum_{i=1}^N |m_i|^-, \quad \bar{m} = \mathcal{U} - \mathcal{D}.$$

Decomposing aggregate mobility: Net mobility and mobility concentration

$$\mathcal{M}_N = \underbrace{\bar{m}}_{\text{Net Mobility}} \times \underbrace{\frac{\text{sign}(\mathcal{S}_\gamma)}{\text{sign}(\bar{m})}}_{\gamma\text{-Sign Correction}} \times \underbrace{\bar{\mathcal{C}}(M)}_{\text{Net Mobility Concentration}}$$

$$\mathcal{S}_\gamma(M) \equiv \sum_{i=1}^N \text{sign}(m_i) |m_i|^\gamma \quad \bar{\mathcal{C}}(M) \equiv \left| \left(\frac{\mathcal{U}}{|\bar{m}|} \mathcal{C}_\gamma^+(M) \right)^\gamma - \left(\frac{\mathcal{D}}{|\bar{m}|} \mathcal{C}_\gamma^-(M) \right)^\gamma \right|^{1/\gamma}.$$

Average Symmetric Mobility

Define average upward and downward signed mobility by $\mathcal{U}^s$ and $\mathcal{D}^s$.

$$\overline{m}^a = \frac{1}{2} \left| \mathcal{G}(Y^P) - \mathcal{G}(Y^K) \right| + 2 \left[\mathcal{D}^s \mathbb{1}\{\mathcal{G}(Y^P) \geq \mathcal{G}(Y^K)\} + \mathcal{U}^s \mathbb{1}\{\mathcal{G}(Y^P) < \mathcal{G}(Y^K)\} \right]$$

- First term: absolute net vertical mobility
- Remaining term: positional reshuffling not summarized by the Gini
- If inequality falls, total upward mobility equals the net gain plus the downward movements that it offsets

Progressive Taxation and Rank Dependence

Under Pareto incomes and an FGM copula for parental and child ranks,

$$C_{\theta}(u, v) = uv [1 + \theta(1 - u)(1 - v)], \quad \theta \in [-1, 1].$$

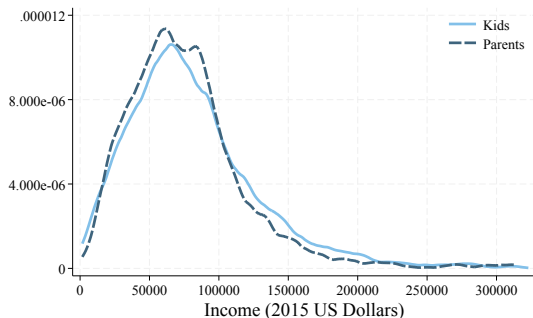
Let $p_G = \alpha_G / (\alpha_G - 1 + \tau_G)$ denote after-tax Lorenz curvature. Then

$$\mathcal{M}_N \propto \frac{p_K}{p_K + 1} + \frac{p_P}{p_P + 1} - 2 \int_0^1 C_{\theta}(1 - t^{p_K}, 1 - t^{p_P}) dt.$$

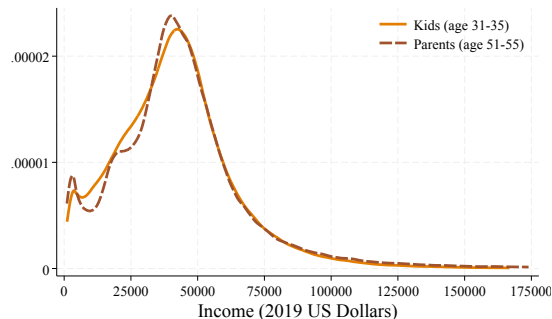
- The effect of common progressivity is affine in rank dependence θ
- Progressivity raises symmetric mobility only below a positive dependence threshold
- With stronger dependence, horizontal mobility fades and inequality compression dominates

Income Distributions Across Generations

United States



Norway



- US children have 7.8% higher average income than their parents
- Norwegian children have 7.6% lower average income than their parents
- Lorenz mobility removes these changes in aggregate income scale

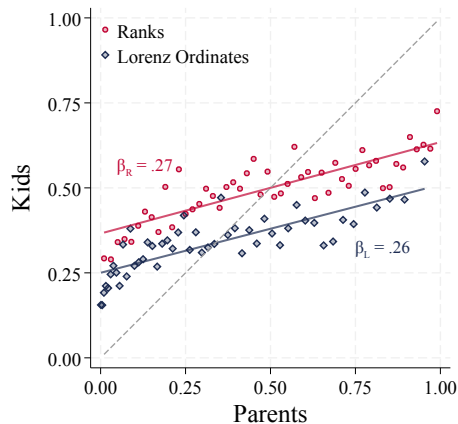
Mobility in Disposable Income: Norway

	Signed mobility			Symmetric mobility		
	$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$	$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$
Lorenz	0.031	0.002	0.129	0.277	0.232	0.350
Rank	—	0.000	−0.036	0.298	0.250	0.374

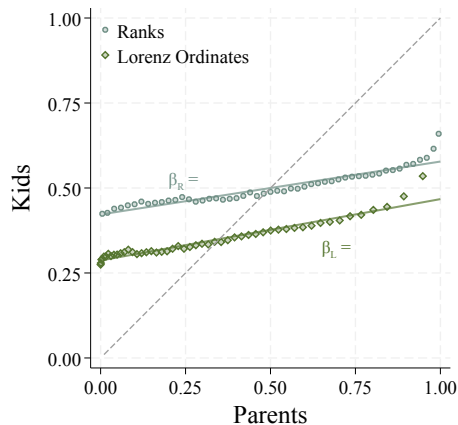
- Taxes and transfers increase signed Lorenz mobility relative to market income
- Symmetric Lorenz and rank mobility are largely unchanged

Intergenerational Regressions: Levels

United States



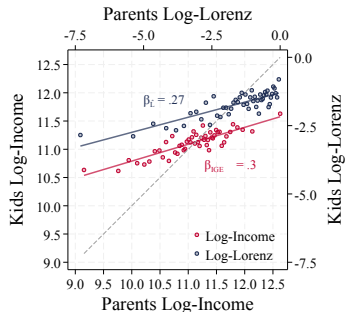
Norway



Binned means and linear fits for ranks and Lorenz ordinates across generations.

Intergenerational Regressions: Logs

United States



Norway

Norway log-income and log-Lorenz
figure pending

Log-Lorenz and log-income regressions use separate axes; their coefficients need not coincide